

WEEK 07 · UX/UI · PROJECT KICKOFF

THE PAGE THAT WORKS

UX, accessibility, and the seven weeks that start today.

WHERE WE ARE

Week 7 of 12. Six weeks of making pages render.
Today: making one work.

THE NEXT 180 MINUTES

Five cycles, you typing in every one: three green checks · a table that becomes cards · tokens that pass · a plan with Claude · what the page announces. Break at 90, assignment at 145.

WHAT THIS UNLOCKS

Your project. You leave this room with a locked topic and a plan you corrected.

“A designer knows he has achieved perfection not when there is nothing left to add, but when there is nothing left to take away.”

ANTOINE DE SAINT-EXUPÉRY – THE OPENING OF LESSON 7, KEPT

TARGET · MINUTE 0

THIS IS WHERE YOU WILL BE AT MINUTE 180

the class list of the community-centre page, rebuilt – one of two right answers

reload

open

החוגים

עשרים ושניים חוגים בחמישה ערבים בשבוע, מיוגה למתחילים ועד גיטרה קלאסית. ההרשמה נפתחת בתחילת כל סמסטר והמקומות מוקצים לפי סדר ההגעה.

לוח החוגים

גיטרה קלאסית

יום שני
שעה 17:30
מדריך אורי שגב
מלא - רשימת המתנה

יוגה למתחילים

יום ראשון
שעה 18:00
מדריך נועה ברק
מקומות פנויים

מחשבים לגמלאים

ציור ורישום

החוגים

עשרים ושניים חוגים בחמישה ערבים בשבוע, מיוגה למתחילים ועד גיטרה קלאסית. ההרשמה נפתחת בתחילת כל סמסטר והמקומות מוקצים לפי סדר ההגעה.

לוח החוגים

יום שני	יום ראשון	יום שלישי
שעה 17:30 מדריך אורי שגב מלא - רשימת המתנה	שעה 18:00 מדריך נועה ברק מקומות פנויים	שעה 19:00 מדריך אורי שגב מלא - רשימת המתנה

targets/week-07.png · 1280

החוגים

עשרים ושניים חוגים בחמישה ערבים בשבוע, מיוגה למתחילים ועד גיטרה קלאסית. ההרשמה נפתחת בתחילת כל סמסטר והמקומות מוקצים לפי סדר ההגעה.

375 · one column, nothing overflows

► **THE POINT** — three things leave this room: an **AUDIT** of a page every tool calls fine, **THIS REGION REBUILT** in Tailwind (cards here; a table that scrolls inside its own box earns the same), and **YOUR PROJECT PLAN**, drafted with Claude and corrected by you. You do each once in a cycle first.

AI THIS WEEK

ASSISTANT. IT DRAFTS; YOU CUT, AND SAY WHY.

LAYER · WEEKS 6-7 · ASSISTANT

ALLOWED

A first draft of your `PROJECT_PLAN.md` (cycle 4). Two class lists for a badge, and you pick one (cycle 2). Why 4.5 and not 3. Then you check it — in the brief, in the colour picker, in your own browser.

NOT ALLOWED

Pasting the bad page and asking for the list of faults — that is the audit, and its list has no severity and no "what breaks". Keeping a draft you did not cut. **Every rejection gets a sentence.**

NO SUBSCRIPTION?

Everything works in the free chat, and nothing in the course depends on a paid plan. Cycle 4 is text in, text out.

CYCLE 1 · 25 MIN

THREE GREEN CHECKS, AND THE MOUSE PUT DOWN

A library page that passes the validator, the console and two widths, and gets one rule from axe. Write down what the tools said — then what they could not see: a status column with no words, a Tab path with no ring, a table 576 wide at 375. Nobody fixes anything.

1 I DO

10 MIN

3 on the next four slides, 7 in the editor · live-code/01-three-green-checks · npx html-validate@9 · console · 768/1280 · axe · checks.txt · then Tab, then 375

2 WE DO

5 MIN

you run the same checks on your copy · five findings in your audit-he.md, each "what breaks" starting with a person · compare with checkpoint 1

3 YOU DO

8 MIN

lab/warmup/ — six faults in one card, find and fix them, the referee says 0 of 6

4 DEBRIEF

2 MIN

"the contrast is low so it is hard to read" is the description twice · ↗ reference

YOU DO · AI-FREE

HERE IS A FINISHED PAGE

a community centre, five weeks of skills applied

reload

open

בית חוגים צור קשר כאן

מרכז קהילתי אלמוגים

המרכז הקהילתי אלמוגים פועל בשכונה מזה שנים רבות ומציע מגוון רחב של חוגים לכל הגילים, מגן הילדים ועד לגמלאים, בתחומי הספורט, האמנות, המוזיקה, המחשבים והעשרה כללית, כאשר כל החוגים מונחים על ידי מדריכים מוסמכים ובעלי ניסיון רב שנים בתחומם, והרשמה לחוגים נפתחת בתחילת כל סמסטר ונמשכת עד למילוי המקומות, כך שמומלץ להירשם מוקדם כדי להבטיח מקום בחוג המבוקש, ובמידה שחוג מסוים מתמלא ניתן להצטרף לרשימת המתנה ולקבל עדכון כאשר מתפנה מקום.

לפרטים נוספים על תנאי ההרשמה, הביטולים וההחזרים לחצו כאן.

החוגים

רשימת החוגים

החוג	יום	שעה	מדריך	מקומות
יוגה למתחילים	ראשון	18:00	נועה ברק	●
גיטרה קלאסית	שני	17:30	אורי שגב	●
ציור ורישום	שלישי	19:00	תמר לוי	●
מחשבים לגמלאים	רביעי	10:00	דני אמסלם	●
כדורסל נוער	חמישי	17:00	רון כספי	●

גלריה



► **LOOK, DON'T EXPLAIN.** Give the room twenty seconds. Then ask what is wrong with it — and write the answers on the board.

NOW RUN EVERY CHECK YOU OWN

THE VALIDATOR

```
$ npx html-validate@9 bad-page.html
$ echo $?
0
```

THE CONSOLE

```
0 errors
0 warnings
0 failed requests
```

VALID HTML5

One `h1`. No skipped heading levels. An `alt` on every image. A real `label[for]` on every field. Scoped table headers.

RENDERS EVERYWHERE

Nothing missing, nothing overlapping, nothing off-screen at 768 or 1280.

No validator has an opinion about whether a page is **usable**.

Every tool you own checks that the page is well-formed. None of them checks that a person can get anything done with it.

PUT THE MOUSE DOWN

Do the main task with the **keyboard only**.

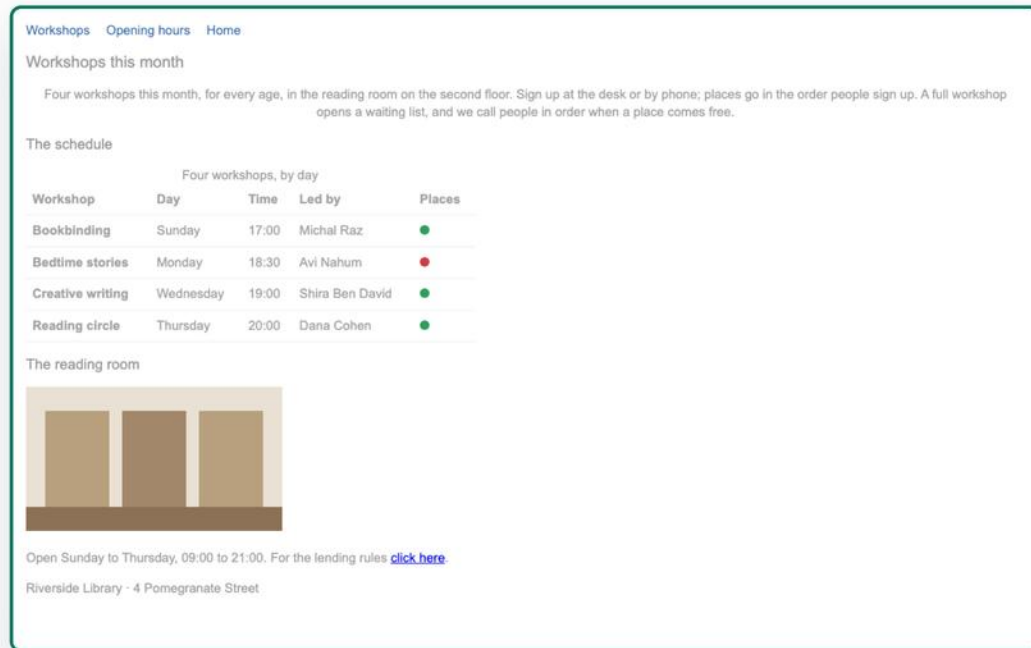
Push the mouse to the side of the desk and complete the thing your page is for. Where you get stuck is the finding. This takes ninety seconds and finds more than any tool.

WHO THIS IS FOR, AND IT IS NOT A MINORITY

Someone with a broken wrist. Someone on a train. Someone who is faster at keyboards than you are. Someone using a screen reader. The last group is the one people think of and the smallest of the four.

CHECKPOINT 1

YOUR SCREEN SHOULD LOOK LIKE THIS — AND THAT IS THE POINT



Compare — the page did not move a pixel. What changed is next to it: `live-code/01-three-green-checks/end/checks.txt` (what each tool said, and what was measured when they were silent) and your own `audit-he.md`, five rows, three groups, every "what breaks" starting with a person.

CYCLE 2 · 25 MIN

HIERARCHY, RHYTHM, AND A TABLE THAT BECOMES CARDS

The library's schedule, lifted out without its stylesheet, rebuilt in utilities: a heading that is measurably first, a table that "fits" at 375 and is worse for it, and cards whose gap between is three times the gap within.

1 I DO

10 MIN

3 on the next three slides, 7 in the editor · live-code/02-table-to-cards · text-3xl font-bold · w-full on the table (wrong) · ul of cards · gap-2 inside, gap-6 between · sm:grid-cols-2

2 WE DO

5 MIN

you type steps 1, 3, 4 · compare with checkpoint 2 · 375: one column, no scrollbar

3 YOU DO

8 MIN

a fourth field and a "New" tag, the rhythm still holds · AI as assistant: two class lists, pick one, one line on why

4 DEBRIEF

2 MIN

the scrollbar went away and the page got worse · ↗ reference

YOU DO · AI AS ASSISTANT

HIERARCHY IS AN ORDERING, NOT A SIZE

- What it is** What the reader sees first, second, third — decided by you, not by the markup order.
- Four levers** **Size · weight · colour · position.** Nothing else.
- The failure** Two levers pointing opposite ways. They cancel, and the page reads bottom-up.
- The test** Look for five seconds, close your eyes. What did you keep?

THE ONE THAT SURPRISES PEOPLE

Size is the weakest of the four. A big pale heading loses to small black body text every time.

GROUPING COSTS NOTHING BUT SPACE

eight fields · no boxes, no headings, no colour

reload open

EIGHT FIELDS, NO BOXES, NO HEADINGS

between : inside 1.0 ☐ add boxes instead how many groups?

Full name

Date of birth

Identity number

Street and number

City

Postcode

Email

Phone

groups a reader will see

?

inside a group 0px

between groups 0px

ratio 1.00

borders drawn 0

One rhythm, so there is no grouping.
Eight fields in a single undifferentiated column — and nothing on this panel is wrong, misaligned or missing.

► **SHOW OF HANDS AT RATIO 1.0:** who sees three groups? Count them. Then drag to 2.0 and ask again.

RHYTHM IS A PROPERTY OF THE SET

CHOSEN PER ELEMENT

```
.card    { padding: 13px; }  
.panel   { padding: 18px; }  
.sidebar { padding: 17px; }  
.hero    { padding: 22px; }
```

TAKEN FROM A SET

```
/* --spacing: 0.25rem = 4 8 12 16 24 32 48 */  
.card    { padding: var(--space-4); }  
.panel   { padding: var(--space-5); }  
.sidebar { padding: var(--space-5); }  
.hero    { padding: var(--space-8); }
```

Nobody can see that 17 is wrong. Everybody can see that a page of 13, 18, 17 and 22 has no rhythm — and neither statement is about any one element.

CHECKPOINT 2

YOUR SCREEN SHOULD LOOK LIKE THIS



Compare — `live-code/02-table-to-cards/checkpoint.png` at 1280; `end/` is the code behind it. `h1` 30px bold against a 16px paragraph · gap 24 between cards, 8 inside · at 375 one card per row and no scrollbar. The dots still say nothing — on purpose, until cycle 3.

CYCLE 3 · 30 MIN

TOKENS THAT PASS, AND A STATUS THAT READS

Eleven colours in @theme and not one says whether it is for text. The dot becomes a pill with a word, every text pair is checked as a number, a quiet grey turns out to be a line colour at 1.40, and the edge of a control is measured against 3.

1 I DO

10 MIN

3 on the next three slides, 7 in the editor · live-code/03-tokens-that-pass · the pill: 6.52 and 6.67 · text-rule on the labels (wrong): 1.40, axe ×12 · text-print-soft: 6.55 · border-edge 4.74 vs border-rule 1.52

2 WE DO

8 MIN

you type steps 1, 3, 4 and write down every ratio · compare with checkpoint 3 · Achromatopsia: every pill still says what it is

3 YOU DO

10 MIN

a fifth card, full, and a "last updated" line · every new pair measured and noted · no text-rule, no brackets

4 DEBRIEF

2 MIN

choose a token by what it is for, not by how it looks · ↗ reference

YOU DO · AI-FREE

A PALETTE FROM ONE NUMBER

one hue · every pair measured off the rendered pixels

reloadopen

hue 268saturation 70%lightness shift 0

SIX TOKENS, ONE HUE

--color-paper
rgb(252, 252, 253)

--color-surface
rgb(245, 242, 248)

--color-ink
rgb(30, 22, 40)

--color-ink-muted
rgb(83, 72, 96)

--color-brand
rgb(92, 29, 165)

--color-control
rgb(111, 89, 136)

EVERY PAIR, MEASURED OFF THE RENDERED PIXELS

text needs 4.5:1 · large text and the edge of a control need 3:1

foreground	on background	ratio	needs	verdict
ink (body text)	paper	17.06:1	4.5:1	pass
ink (text on a card)	surface	15.77:1	4.5:1	pass
ink-muted (secondary text)	paper	8.31:1	4.5:1	pass
ink-muted (secondary on a card)	surface	7.69:1	4.5:1	pass
brand (a link)	paper	9.42:1	4.5:1	pass
brand (a link on a card)	surface	8.71:1	4.5:1	pass
paper (text on a button)	brand	9.42:1	4.5:1	pass
control (the edge of a field)	paper	5.92:1	3:1	pass
control (a field on a card)	surface	5.47:1	3:1	pass

All nine pairs pass, at every hue on the slider. Try to find a hue that breaks it — you will not, because hue is not the variable that carries contrast. Now move the lightness shift by one.

► **BEFORE CYCLE 3.** Ask the room for a hue. Type it in. Then ask them to find a hue where the palette fails.

TWO THRESHOLDS, AND THE ONE NOBODY KNOWS

4.5 : 1

Normal text. This is the one everybody has heard of.

3 : 1

Large text — 24px, or 18.66px bold (18pt, 14pt bold).

3 : 1, NON-TEXT

The edge of a control. The border of a text field, the outline of a button — anything you need to see in order to find the thing.

A page can pass a contrast checker on every single word and still have input fields nobody can locate. **The checker looked at text.**

COLOUR IS NEVER THE ONLY CUE

A COLOURED DOT

Green for available, red for full. In greyscale the column is empty; to a screen reader it was always empty.

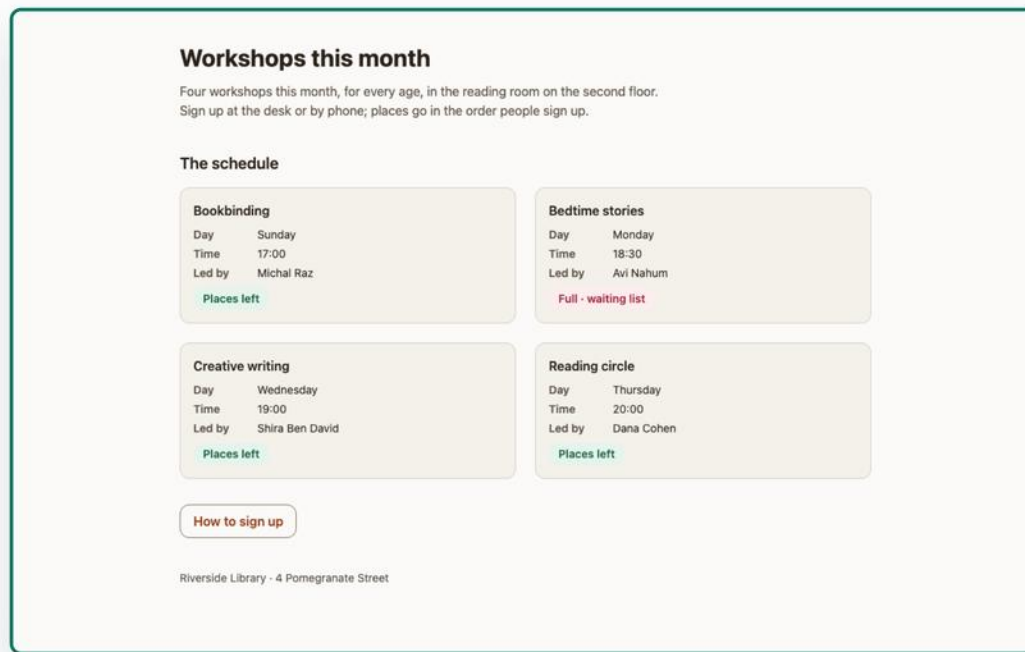
COLOUR *AND* A WORD

A green dot beside the text “available”. The colour is the fast path, not the only path.

► **WCAG 1.4.1, LEVEL A.** Cycle 3 turns the dot into a pill that says **PLACES LEFT** — and reads it again in greyscale.

CHECKPOINT 3

YOUR SCREEN SHOULD LOOK LIKE THIS



Compare — `live-code/03-tokens-that-pass/checkpoint.png` . Every pill says a word · labels at 6.55, pills at 6.52 and 6.67 · the control edge at 4.74, the card edge at 1.52 and that is fine: a card is not a control. Then Rendering, Achromatopsia.

BREAK

10 MINUTES

Back at 100. Cycle 4 is the AI cycle — a chat window open, free or paid, and the four lines about your topic ready to paste. No topic yet? Pick two candidates from `project-topics-he.md` now.

CYCLE 4 · 25 MIN

A PLAN WITH CLAUDE, AND THE THREE LINES YOU WRITE

Four lines about a topic and a list of ten sections: the draft comes back complete, typed, tidy — and plans a different project, with accounts and a server. You count, search and open the API in a browser, cut it to the brief, and write what it guessed, what you fixed, and why.

1 I DO

8 MIN

3 on the next two slides, 5 in the chat · ai-lab/04-plan-with-claude · P1 live · count the core · search "server" · 11 of 14 on the automatic checks · the good example on screen

2 WE DO

6 MIN

P1 in your own chat, with your topic · find what the brief forbids before it gets a name

3 YOU DO

9 MIN

your own P2 with the whole template and the brief's constraints · open the API in a browser · cut · write section 11

4 DEBRIEF

2 MIN

a complete draft is not a correct one · ↗ reference

YOU DO · AI AS ASSISTANT

SECOND HALF · THE KICKOFF

YOUR PROJECT

Fifty per cent of the grade. Six weeks, this one included. It starts now, and it is built **during** the course rather than after it.

WHAT YOU BUILD

One application that manages a collection. Three pages, one entity.

WHAT YOU LEAVE WITH TODAY

A locked topic and a committed
`PROJECT_PLAN.md`.

WHAT VERIFIES IT

Week 13. In the lab, no internet, no AI, on your own code.

THE LOOP, AND IT STARTS WITH A DOCUMENT

1**Talk**

describe the topic. Ask for no code.

2**Plan**

it fills `PROJECT_PLAN.md`.

3**Review**

you fix what it guessed.

4**Commit**

and the plan is now the contract.

Reviewing a paragraph in Hebrew takes a minute. Reviewing eighty lines of code takes half an hour — **and nobody pays for that half hour**, which is why unreviewed code ships. The document is where the review is cheap.

IT WILL BE WRONG, AND THE FIRST WAY IS THE SIZE

TWELVE FEATURES, A LOGIN, A SERVER

Ask for "a complete plan" and the draft plans a different project: accounts, sync through a backend, sharing with family.

Count the core. Search for "server". It still scored 11 of 14 on the automatic plan checks.

AN ENTITY TOO BIG

Three tables called one. Section 3 is where you catch it.

A MISSING STATE

“Not yet searched” and “found nothing” are two different empties.

AN API THAT NEEDS A KEY

`perenual.com` answers
`400 Missing API Key`. **The only evidence is JSON you saw in your own browser.**

CYCLE 4 · ON SCREEN BEFORE ANYONE WRITES

WHAT A CORRECTED PLAN LOOKS LIKE

SECTION 3 · ONE ENTITY, TYPED

`title: string · year: number | null ·`
`rating: 0..5 · addedAt: string`. A field that can be missing says so; a notation more exact than `number` is welcome.

SECTION 11 · THREE LINES

What Claude guessed — twelve features, a login, "a small Node server", three booleans. **What I fixed** — five in the core; no login, no server; one `status`. **Why** — the brief says `localStorage` only, **and I checked it there** before deleting.

► **AI-EXEMPLARS/EXEMPLARS/PROJECT-PLAN-GOOD.MD** in your course materials. Guess · fix · reason — and the reason holds a check you could repeat. "I went over it and fixed it" is not a line.

THERE ARE TWO EMPTIES, AND THEY ARE NOT THE SAME SENTENCE

ONE PANEL FOR BOTH

“No results found.” — shown before the user has typed anything. The application reads as broken on arrival.

NOTHING SEARCHED YET

“Search for a class to begin. Type a name or an instructor above.”

SEARCHED, FOUND NOTHING

“No classes found for **trapeze**. Try a different spelling, or search by instructor.”

An empty state that does not say **what to do next** is a blank rectangle with a sentence on it. And in section 7 of your plan the four states are **one field with four values** —

`status: 'idle' | 'loading' | 'error' | 'success'` — not three booleans.

NO SUBSCRIPTION? THE SAME LOOP.

THE WHOLE LOOP IS TEXT

Paste the template and the brief's constraints into the free chat, save what comes back, cut it, write section 11. Nothing in cycle 4 needs anything else.

THE LAYER THIS WEEK

Assistant. It drafts; you choose, cut, and **say in writing** what you rejected and why. That sentence is the graded part.

THE GRADED PART IS IDENTICAL

Section 11 does not care which chat the draft came from. **Nobody is blocked by not owning a subscription.**

CHECKPOINT 4 · THREE QUESTIONS

BEFORE YOU LEAVE CYCLE 4

- 1 Did you count the core in your P1 answer, and search it for "server"?**
Not "all ten sections are full".
- 2 Does your P2 carry constraints you can count or search — and did you open its API in your own browser?**
A 400 with "Missing API Key" is an answer too.
- 3 Does the "why" line in your section 11 contain a check?**
What you opened, what came back, which part of the brief.

CYCLE 5 · 20 MIN

THE MOUSE PUT DOWN AGAIN, AND WHAT THE PAGE ANNOUNCES

The rebuilt region, met the way a keyboard and a screen reader meet it: a ring written in utilities, a link whose name says where, an alt that is a decision, and a line that will be announced when week 9 changes it. All from weeks 1–3; the spelling is new.

1 I DO

7 MIN

2 on the next slide, 5 in the editor ·
live-code/05-what-it-announces ·
focus:outline-none (wrong) · focus-
visible:outline-2 · "click here" to
"the lending rules" · alt="" ·
role="status"

2 WE DO

5 MIN

you type steps 2–5, mouse aside ·
compare with checkpoint 5 — by
Tab, not by eye

3 YOU DO

6 MIN

a "Details" link on every card, each
with its own name · AI as assistant:
three ways, pick one, one line on
why

4 DEBRIEF

2 MIN

we removed the keyboard's ring to
hide a mouse ring that was never
there · ↗ reference

WHAT A SCREEN READER WOULD SAY

role + name + state, computed from the DOM, live↻ reloadopen ↗

☐ label after the field ☐ aria-label on the Send button ☐ the Cancel control is a div ☐ the Delete button is icon-only

THE PANEL — TAB THROUGH IT

Full name

Class

Beginner yoga

☐ I accept the registration terms

Send

Cancel

Delete registration

WHAT A SCREEN READER WOULD SAY

ANNOUNCEMENT LOG

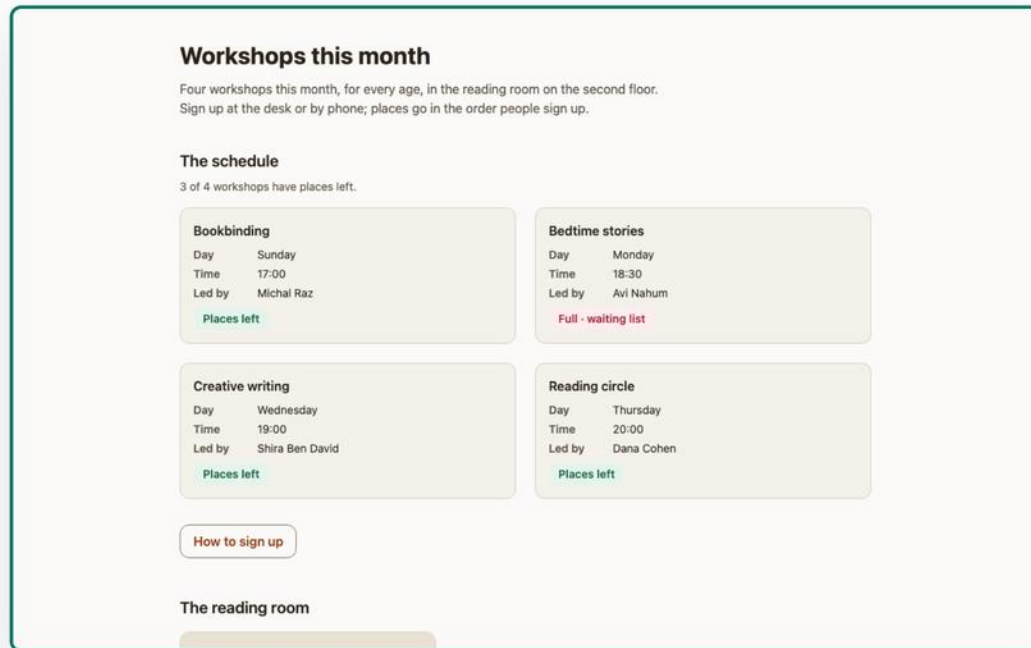
Press Tab on the panel. Each stop is announced here, computed from the DOM.

A stand-in, not a screen reader. It implements the useful part of the accessible-name computation and the states that matter here — enough to make the link between an attribute and a spoken sentence visible. Run the real thing too.

► **TURN ON “ARIA-LABEL ON THE SEND BUTTON”.** Read the visible text aloud, then read the announcement aloud. Ask which is right.

CHECKPOINT 5

YOUR SCREEN SHOULD LOOK LIKE THIS — AND TAB SHOULD SAY MORE



Compare by keyboard — a solid 2px ring on How to sign up and on the lending rules, none on a mouse click. In the accessibility tree: `status`, two links with names that say where, and no `img` called "image".

ASSIGNMENT · 145–175

CLASS ASSIGNMENT 7: ASSEMBLE, THEN EXTEND

1 Audit · cycle 1

the community-centre page, against the fifty-five-row checklist. Sixteen faults; twelve findings is the floor.

2 Redesign · cycles 2, 3, 5

its class list, in Tailwind: a structure, not a width · a pill that says the word · a token that passes · a ring and a name.

3 Plan · cycle 4

your own `PROJECT_PLAN.md`, drafted with Claude, cut by you, section 11 written, committed and pushed.

NO JAVASCRIPT THIS WEEK

The redesign is markup and utilities. Focus management and validation timing are things you can now **recognise and audit**; building them is weeks 8 and 9.

THE AUDIT AND THE PLAN ARE WEIGHTED THE SAME

Thirty-five points each. Finding sixteen faults and writing no plan scores the same as the reverse.

► **DUE BEFORE LESSON 8** — repo URL and SHA in Moodle.

Lock your topic **today**.

Twenty-two in the pool, or your own. Four questions to pass. I come to your table before 175 — say one topic and its one entity.

CHOOSING A TOPIC, IN FOUR QUESTIONS

- 1 Is there a collection the user manages?**
add, edit, delete, search, filter. A game with no collection does not fit.
- 2 Is the entity one thing?**
if you need three tables, pick one of them.
- 3 Is there a keyless public API?**
verify it in a browser. Not in `curl`, and not because Claude said so.
- 4 Is there a third page that is entirely derived?**
a summary, a breakdown. Not “About”.

Twenty-two vetted topics in `project-topics-he.md`, each with its entity fields, a verified API and the one risk that actually sinks it. Your own topic is welcome — **approved today, not next week.**



THREE QUESTIONS BEFORE YOU GO

A page passes the validator, the console is clean, and it renders at all three widths. What

1 have you learned about it?

That it is well-formed. Nothing about whether anyone can use it.

A region shows data. How many states does it need, and which one is usually missing?

2

Four. Usually both empties collapsed into one, or no error state at all.

A plan has ten full sections and scores 11 of 14 on the automatic checks. What have you

3 learned about it?

That it looks like a plan. Count the core, search for "server", open the API in a browser.

GO DEEPER

The reference deck for this week — every slide the lecture pointed at, in order.

[2 · Contrast: three numbers, and what each covers](#) ← spine 15

[3 · The four states, and how each one fails](#) ← spine 21

[7 · The keyboard contract](#) ← spine 4

[9 · Does this need ARIA?](#) ← spine 29

[10 · Alt text, as a decision](#) ← spine 29

[11 · The three scales](#) ← spine 10

[12 · Measure, and the trap in ch](#) ← spine 10

[13 · Testing it on a person](#) ← spine 4

[14 · The five misconceptions](#) ← spine 35

[15 · Where to look, in order](#) ← spine 4

[17 · The same words, five ways](#) ← spine 10

[19 · Layout, and the length of a line](#) ← spine 10

[23 · A type scale, and one size that does not belong](#) ← spine 10

[25 · The same region, badly and well](#) ← spine 21

[30 · The focus ring is not decoration](#) ← spine 29

[31 · Where does focus actually go?](#) ← spine 4

[33 · The accessible name](#) ← spine 29

[35 · Alt text that says something](#) ← spine 29

[37 · Where the points are](#) ← spine 21

[38 · A finished one, running](#) ← spine 21

[40 · Six weeks, and how to know you are behind](#) ← spine 21

[42 · Colour alone: WCAG 1.4.1, and why axe cannot see it](#) ← spine 15

[43 · Tailwind v4: outline-none, outline-hidden, focus:, focus-visible:](#) ← spine 29

[44 · role="status": a live region you declare before it changes](#) ← spine 29

[45 · What the automatic plan checks can and cannot see](#) ← spine 21